

GLOFeagles 2026 Challenge Evaluation Report

Team Name: CyberCorps

1. Objective

The objective of this evaluation was to assess the performance of the proposed DeepLabV3+ based semantic segmentation model for glacier-related environmental feature identification and GLOF risk assessment.

The model was evaluated on a validation dataset derived from the provided labeled subset.

2. Evaluation Dataset

The labeled dataset consisted of 60 annotated samples.

Data Split

Dataset	Number of Images
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Training Set	48
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Validation Set	12
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The validation set was used to measure segmentation performance and model generalization.

3. Evaluation Metrics

The following metrics were used:

Mean Intersection over Union (mIoU)

Measures the overlap between predicted segmentation masks and ground truth masks.

Overall Accuracy

Measures the percentage of correctly classified pixels.

Precision

Measures the proportion of positive predictions that were correct.

Recall

Measures the proportion of actual positive pixels correctly identified.

F1 Score

Harmonic mean of Precision and Recall.

Cohen's Kappa

Measures agreement between predicted labels and ground truth while accounting for random agreement.

4. Validation Results

Overall Performance

Metric	Value
mIoU	0.5388
Overall Accuracy	95.22%
Macro Precision	0.2417
Macro Recall	0.1895
Macro F1 Score	0.1998
Cohen's Kappa	0.2062

The best model checkpoint was automatically selected using validation mIoU and early stopping.

5. Class-wise Performance

Class	IoU	Precision	Recall	F1 Score
Background	0.9519	0.9587	0.9925	0.9754
Glacial Lake	0.1257	0.4917	0.1445	0.2234

Certain classes showed limited representation within the labeled subset, which influenced class-wise metric stability.

6. Qualitative Evaluation

Visual inspection of generated segmentation masks indicates that the model successfully identifies major glacial lake regions and background areas.

Uncertainty maps generated using Monte Carlo Dropout highlight regions where the model exhibits lower confidence, providing additional interpretability.

The GLOF risk assessment module successfully generated risk maps and per-image JSON reports for all challenge images.

7. Output Generation

The final model was used to generate outputs for all 575 challenge images.

Generated outputs included:

- Segmentation masks
- Visualization masks
- Uncertainty maps
- Risk maps
- Per-image risk reports

The inference pipeline successfully processed the complete challenge dataset using the saved model checkpoint.

8. Conclusion

The evaluation demonstrates that the proposed DeepLabV3+ based segmentation framework is capable of identifying glacial lake regions and generating supporting uncertainty and risk assessment outputs.

The model achieved a best validation mIoU of 0.5388 and an overall pixel accuracy of 95.22%, while successfully producing outputs for all 575 challenge images.